



ECOCYCLE

Smart Waste, Clean Cities

A real, full-stack digital ecosystem connecting citizens, collectors & city authorities — for a cleaner, circular Kampala.

SDG 11

SDG 12

SDG 13

THE PROBLEM

Kampala is drowning in its own waste

2,500t

waste generated every day

~48%

never collected

35+

killed, Kiteezi collapse 2024

On **9 August 2024**, heavy rain collapsed the overfull Kiteezi landfill — burying homes and killing dozens. It had been past capacity since **2012**, yet still took 1,500–2,000 tonnes a day.

ROOT CAUSE

It's a broken information loop

Citizens can't report dumps. Collectors work blind, with no job lists or proof. Cities have no real-time data. And nobody is rewarded for recycling — so recyclables get buried.

EcoCycle reconnects all three.

THE SOLUTION

One app, three roles, one loop



Citizen

Report waste with photo + GPS, get AI sorting, schedule & pay for pickups, earn rewards.



Collector

Live job queue, navigation, photo proof-of-collection, live truck location.



City / Admin

Live map of reports, emergency alerts, and an SDG impact dashboard.

THE INTELLIGENCE

AI that sorts waste — on your phone

Snap a photo. TensorFlow.js + MobileNet classify it instantly:



Organic

banana peel, food waste



Plastic

bottles, packaging



E-waste

phones, electronics

On-device = free to scale, privacy-preserving, and works **offline** — built for Kampala's connectivity.

IMPACT

Built around three UN SDGs



SDG 11.6

Reduce cities' environmental impact via better municipal waste management.



SDG 12.5

Cut waste through recycling & reuse — driven by AI sorting + rewards.



SDG 13.3

Climate action: divert organics to cut landfill methane; build climate literacy.

UNDER THE HOOD

A real full-stack app — not a mockup

```
PWA (installable · offline · on-device AI)
  | HTTPS + JWT
Node + Express — REST API (reports, jobs, rewards, pay...)
  |
SQLite database      Integrations (SMS · Email ·
(users, reports,    Flutterwave pay · Google)
points, jobs)       live ↔ sandbox via .env
```

JWT + bcrypt auth · OTP · Google Sign-In · Leaflet maps · Chart.js · Service-worker offline

WHY WE WIN

Others solve pieces. We close the loop.

Capability	Yo-Waste	Homeklin	T-Bin	EcoCycle
On-demand pickup + pay	✓	✓	—	✓
Citizen dump reporting	—	—	—	✓
On-device AI sorting	—	—	~	✓
Gamified rewards	—	—	—	✓
City SDG dashboard	—	—	—	✓
Unified 3-role ecosystem	—	—	—	✓

FROM CRISIS TO CIRCULAR

Smart Waste, Clean Cities

A working platform, ready to deploy — turning every citizen's phone into part of the solution.

 github.com/webdev123-tech/ecocycle

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